**Report: Pre Professional Practice**

Faculty of Engineering and Applied Sciences,
Universidad de los Andes

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## Abstract

This report summarizes my pre-professional practice at the Faculty of Engineering and Applied Sciences, Universidad de los Andes, in December 2024–January 2025. I worked in the Environmental Civil Engineering Investigation Department under Professor Sichem Guerrero at the Engineer Investigation and Innovation Center (EIIC). My contribution was to the project "Simultaneous use of polycyclic aromatic compounds and CO as reducing agents for NO reduction under highly oxidizing conditions in a novel calcium and copper-nickel catalyst on a core-shell YCeO2@M (M=TiO2, ZrO2) support," analyzing a three-phase system of isooctane, water, and surfactant using reverse micelles.

My tasks included sample preparation, viscosity and density measurements, and particle size analysis with a Zetasizer. I also modified Python code to visualize results in a three-way plot. Initially, understanding complex concepts and mastering lab techniques was challenging, but guidance from my supervisor helped me adapt.

This experience enhanced my technical skills, deepened my interest in research, and strengthened my critical thinking and problem-solving abilities in an academic setting.

## 1. Introduction

The main objective of the pre-professional practice was to contribute to a study on a three-phase system comprising water, surfactant and isooctane for the synthesizing of nanoparticles via the reverse micelles process. This process involves the formation of aqueous domains dispersed in a continuous oil phase. These domains are important because they provide a confined environment where a variety of reactants can be introduced for control reactions. The research focused on the characterization of this solution so it can be applied  in the synthesis of metal oxide particles, so that through different methods, the solids could be studied.

### 1.1 Company Description

The Universidad de los Andes, founded in 1989 by a group of academics, emphasized on the complete formation of professionals through various courses and degree programs. The university has around 4,000 employees and offers a wide range of majors in all areas. Within the engineering faculty, there are five specialties: computer science, civil engineering, environmental engineering, electrical engineering and industrial engineering.

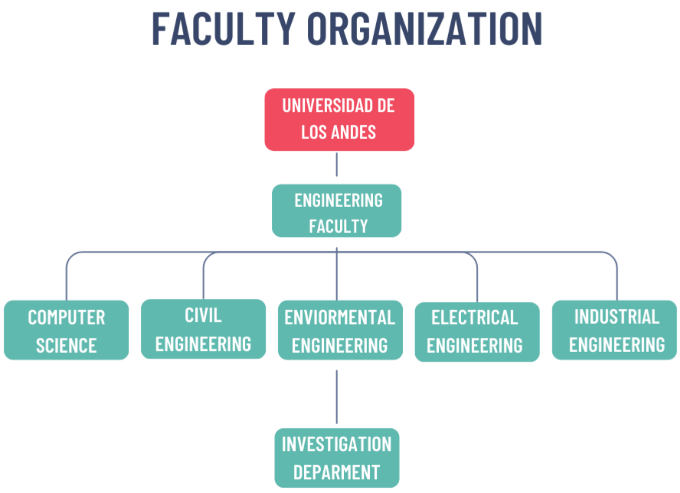



**Diagram 1.1:** Engineer faculty organization
**Source:** Own elaboration

The Faculty of Engineering and Applied Science aspires to form excellent engineers, who are capable of working in different fields and have the ability to perform different technological projects for the development of the country and world. To achieve this they have academics who are capable of developing relevant research while also being able to collaborate with the industry needs.

**Image 1.1** Faculty of Engineering and Applied Science.

### 1.2 Department Description

The pre-professional practice was conducted within the Faculty of Civil Engineering and Sciences, specifically in the environmental engineering department, which is broadly divided into two key areas: the study of physical-chemical processes and the study of microorganisms. My practice was focused on the physical-chemical processes, specifically in research related to catalytic reactions. The investigation department has hosted a wide range of investigations, spanning topics from hydraulic studies to microorganism research. The specific focus of the practice was within the catalysis research area of the Environmental Engineering department.

My direct supervisor was Professor Sichem Guerrero, who led the research project I participated in. He currently has the position of Coordinator of the Environmental Engineer Department and teaches the courses *Machines of Physical and Chemical Transformation* and *Heat Transfer*. Professor Guerrero holds a degree in Chemical Engineering, a master's degree in Chemical Engineering and Engineering Science from the Universidad de Chile, and a doctorate in Engineering Science from the University of Notre Dame, USA.

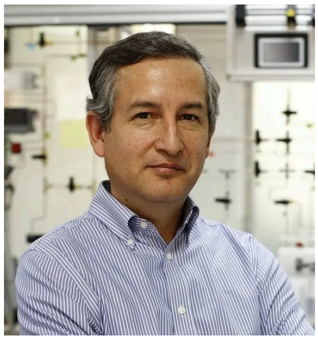


**Image 1.2** Professor Sichem Guerrero

## 2. Work Details

### 2.1 General Activities

During the two months I worked as an academic collaborator for the research project carried out in the Engineering Investigation department.

As previously mentioned, this project aimed to analyze a three-phase system using the reverse micelle process for its potential application in solid formation and catalysis studies. To achieve this, various analyses were conducted, including the characterization of nanoparticle sizes, as well as the density and viscosity of each sample. A total of 18 samples were prepared, each with varying percentages of isooctane, water, and surfactant.

**Understanding of the project**

The initial activity carried out was the understanding of research concepts, for which various materials were studied.

As mentioned earlier, the primary objective of this research was to analyze a three-component system, known as microemulsion systems. These systems are thermodynamically stable and consist of a hydrophobic phase and a hydrophilic phase stabilized by a surfactant. This study focused on reverse microemulsions, which aqueous domains dispersed in a continuous oil phase, referred to as reverse micelles. These systems are notable because they allow the introduction of a variety of reactants for studying controlled reactions.

Another significant aspect of the research was the synthesis of inorganic nanoparticles, which have the potential to produce materials that do not aggregate, enabling the development of materials with unique properties.

The first week of the pre-practice was dedicated to building a theoretical foundation. During this time I spend most of my time in the lab reading different research papers and discussing them with Professor Guerrero to deepen my understanding of the topic and research I was taking part in.

**Sample making**

Another key activity I participated in was sample preparation. Initially, I prepared 6 mL of each sample, followed by a larger batch of 30 mL, with each sample containing different proportions of water, surfactant, and isooctane. To determine these proportions, I used an Excel spreadsheet provided by my professor, which facilitated the calculation of the corresponding volumes in milliliters for each component. The 6 mL samples were later used for nanoparticle size analysis, while the 30 mL samples were designated for viscosity and density measurements.

### Study of particle size


Another of my responsibilities was analyzing the particle size in each sample. For this, we used a Zetasizer, a device that employs the light scattering method to measure micelle size. This technique is based on the interaction between light and particles in suspension, where fluctuations in scattered light intensity are used to determine particle size distribution. Since the machine belongs to the Biomedical Investigation and Innovation Department, I operated it under the supervision of Professor Paulo Díaz. The procedure involved taking 1 mL of each sample, placing it in a small container, and then inserting it into the machine. The Zetasizer then generated an Excel spreadsheet with results from three measurement trials.

Later, I calculated the average of the three measurements for each sample and assigned the corresponding value. Using a code provided by Professor Sichem, I then plotted the data to visualize the results.  For this I had to use my knowledge in coding since I had to adjust the code to achieve it.

### Study of the Viscosity and Density

To measure viscosity, I used an apparatus called an Ostwald viscometer. To ensure proper usage, I watched videos and read various Google sources about it. The viscometer measures viscosity by timing how long the liquid takes to travel from one point to another. It took me approximately two business weeks to measure each sample since I had to wash the viscometer between measurements and wait for it to dry completely.

For measuring the density, I used a pycnometer along with the corresponding equation. This process was quicker than using the viscometer, as the pycnometer dried faster. As with previous tasks, I had to study various sources to fully understand its proper use.

For both density and viscosity measurements, I recorded the obtained data in an Excel spreadsheet that I created. This spreadsheet included the necessary equations to calculate the final values for each sample.

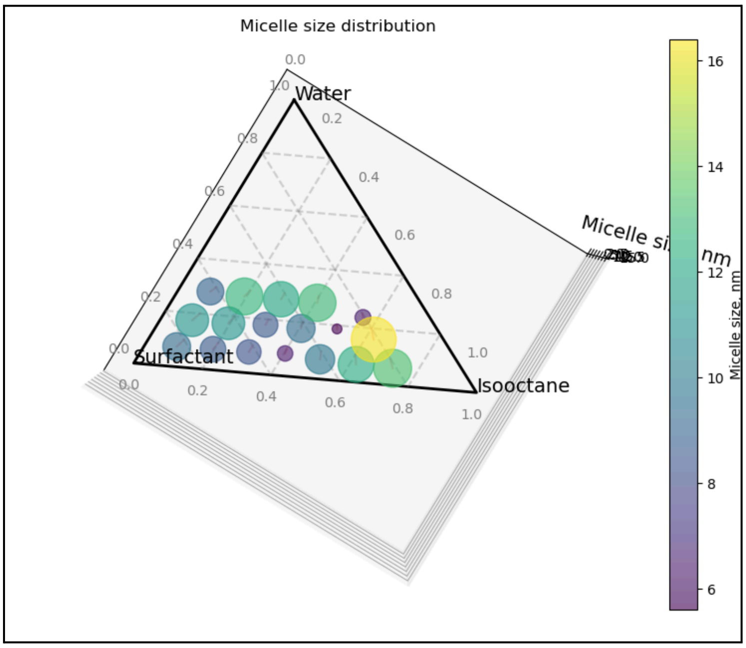
### 2.2 Most Challenging Activity

During my time in the laboratory, I encountered various challenges, primarily because everything was new to me, and I had never worked on a similar investigation before. From sample preparation to data analysis, each step presented unique difficulties that I had to overcome. Given the importance of the research, every procedure had to be carried out with meticulous precision.

The most challenging aspect for me was understanding and analyzing the theoretical framework behind the three-phase system and reverse micelles. As mentioned earlier, I had never studied this area of science before. To overcome this, I dedicated the first week to intensive study and continued researching various topics throughout my two months in the lab. As new tasks were assigned, I had to deepen my understanding of different concepts to successfully complete my work.

When discussing the study and research I conducted, it is important to highlight that one of the key aspects was learning the proper use of Excel and coding. As I mentioned earlier, I used both extensively throughout my work, which required me to dedicate a significant amount of time to mastering them. Understanding how to efficiently organize data, perform calculations, and generate visual representations in Excel, as well as writing and modifying code for data analysis, was a crucial part of my learning process. This not only helped me complete my tasks effectively but also strengthened my technical skills for future research and academic work.

My supervisor, Professor Sichem, played a key role in helping me overcome these challenges. He consistently guided me through the research, recommending various papers to read and suggesting different programs to use. His support was essential in deepening my understanding of the project and refining my skills. Additionally, other researchers working in the laboratory on different projects also contributed to my learning in various ways. These interactions often took place in a more casual manner, fostering an environment of continuous intellectual stimulation. This is something I truly value from my internship, as it not only enhanced my knowledge in specific areas but also encouraged critical thinking and curiosity beyond my immediate research focus.




**Graph 2.1** Micelle size distribution

In Graph 2.1, a three-way plot is displayed, for which I assisted in modifying the code. Although my professor wrote most of it, my task was to adjust a specific part so that the graph accurately displayed the results I had previously calculated. This was still a challenging task, as I had never worked with a graph of this complexity before, nor had I modified code for data visualization in this way.

To ensure I made the correct modifications, I first had to carefully study the existing code to understand its structure and functionality. This involved analyzing how the data was processed, formatted, and ultimately represented in the graph. Only after gaining a solid understanding of the code was I able to make the necessary changes.

Despite the initial difficulties, successfully adapting the graph to display my results was a rewarding experience. It not only strengthened my coding skills but also deepened my understanding of how data analysis and graphical representation are integrated in research.

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## 3. Analysis

### 3.1 Company Analysis

My internship took place in two laboratories at the Universidad de los Andes, in the Biomedic Investigation and Innovation Center (BIIC) and the Engineering Investigation and Innovation Center (EIIC). Both of these labs host a wide range of research projects conducted by students and academics from different fields.

Regarding the EIIC, where I spent most of my time, I noticed that it is highly equipped for various research areas, including microbiological studies and physicochemical processes. The center houses a wide range of specialized equipment, from heat transfer units to ovens used for microorganism cultivation, supporting diverse scientific investigations. It can also be found in smaller equipment like pipets, shackers, etc.

Another notable characteristic of both laboratories is their cleanliness. Students ensure they clean up after themselves, maintaining an organized workspace that allows research to be conducted efficiently. This collective effort fosters a pleasant and professional work environment.

During my internship, the laboratory staff included academics, doctoral students, undergraduate research students, and assistants. While working at the BIIC, I collaborated with Professor Paulo Díaz from the Nutrition and Dietetics Department. However, I spent most of my time at the EIIC, where I worked under the guidance of Professor Sichem. Both professors were vital for me to execute the task successfully.

I had the opportunity to observe the work of doctoral students and title research students, who approached their research with precision and dedication, actively engaging with professors whenever guidance was needed. This highlighted the collaborative and supportive environment within the laboratory, where everyone could work efficiently and comfortably alongside one another.

**Table 1:** Laboratory Organization

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|---|---|---|---|---|---|
| **Principal Investigator** | Sichem Guerrero | Felipe Scott | Alberto Vergara | César Huiliñir | Patricio Moreno |
| **Post Doctoral Researcher** | Blanca Araya | Felipe Carreño | Jorge Vidal | Luz Yañez | Cristian Brito |
| **Phd Students** | Francisco España | Javier Saleh | Sebastián Sepúlveda | Paz Torres | Natalia Fernandez |
| **Master's Students** | | Amparo Melgarejo | Valentina Bila | | |
| **Research Assistants** | | Dana Piras | Ignacia Molina | | |
| **Undergraduate students** | Pía Sepúlveda | Rodrigo Vatcky | Ignacia Gana | Valentinca Cardenas | Antonia Lizana |

In Table 1, the organization of the laboratory of environmental engineering is presented. I chose to display it this way instead of using a diagram because the structure is not hierarchical—there are no direct supervisory roles. Instead, some individuals hold higher academic positions, but their work is independent of the rest. Since each researcher focuses on their own investigation, this format better reflects the collaborative nature of the lab rather than a traditional chain of command.

### 3.2 Work Analysis

My work primarily consisted of assisting Professor Sichem in the “Simultaneous use of polycyclic aromatic compounds and CO as reducing agents for NO reduction under highly oxidizing conditions in a novel calcium and copper-nickel catalyst on a core-shell $YCeO_2@M$ ($M=TiO_2$, $ZrO_2$) support” investigation. As mentioned before, I carried out various activities, including sample preparation, particle size analysis, and viscosity and density studies. I also used Python and Excel to organize the collected data and generate graphs.

The work I carried out had a significant impact on the research area I was involved in, as the study was conducted solely by Professor Sichem and me. This meant that I actively participated in all aspects of the research. Additionally, the professor aims for this system to be used by students working on their theses, which is another important impact to consider.

However, I do not believe I had an impact on the overall organization of the laboratory, as our study was independent of the other ongoing projects, and I was not involved in any of them.

Unfortunately, since I was only in the laboratory for two months, I was unable to see how the research progressed beyond my time there. As a result, I did not have the opportunity to observe the application of the three-phase system, such as the formation of solids.

One of the main mistakes I made, more than once, was mismeasuring the various liquids that make up the three-phase system. This was a significant error, as failing to notice it could have seriously impacted the study’s development and results. This was especially critical since each sample had unique proportions of water, isooctane, and surfactant. Given that only one sample was prepared per set of proportions, any miscalculation in measurements would have compromised all subsequent analyses.

Furthermore, as I mentioned before, I had never participated in a research project of this nature, which made the initial stages somewhat challenging. It took me some time to adapt to the workflow and avoid feeling frustrated by my lack of understanding of certain aspects of the study. However, as I gradually familiarized myself with the key concepts and general objectives of the research, I was able to work more efficiently and with greater confidence. This learning process allowed me to improve my problem-solving skills and become more comfortable with the different techniques and methodologies involved.

## 4. Personal Reflection

As I mentioned previously, at the beginning, I found the internship quite challenging due to the level of knowledge required in an area I had never studied before. Despite this, I recognize my effort to learn and how I gradually gained confidence in the research, eventually reaching a point where I could carry out most of the tasks independently, without the professor's assistance. This experience significantly helped me develop my learning pace and build confidence in my own abilities. Since I was the only student involved in the research, every successfully completed task was a direct reflection of my work—of course, always guided by Professor Sichem's support.

Continuing on this point, I believe this experience was highly enriching from a professional standpoint, as it served as a guide in helping me explore the areas I may want to focus on in the future. My interest in academic research grew, and this internship undoubtedly contributed to my professional development by introducing me to a field of study I had not previously explored. Additionally, it enhanced my ability to learn independently, strengthening my problem-solving skills and self-sufficiency in approaching new challenges.

Another aspect I would like to highlight is the work environment in the laboratory. Since everyone there is involved in academic research, interesting conversations often arose, helping me develop my critical thinking skills and further fueling my curiosity and desire for knowledge. This intellectual exchange made the experience even more enriching, as I was constantly exposed to new ideas and perspectives.